Author: Imama Zubair AI Intern @ NETIXSOL

Week 4 Day 1

Summary

Problem Framing

The goal is to predict whether a person earns more than $50K per year using the Adult Census Income dataset. The model can be used for targeted marketing, where correctly identifying high-income individuals helps reduce wasted outreach(wasted calls etc).

Chosen Metric

The chosen metric is **Precision.** Since the business objective is to avoid contacting the wrong people, it is more important that predicted high-income individuals are actually high-income, even if some are missed.

Baseline Results

Majority Class: Accuracy 0.761, Precision 0.000, Recall 0.000, F1 0.000 (not good at all)

Education Rule: Accuracy 0.753, Precision 0.484, Recall 0.497, F1 0.491

Capital Gain Rule: Accuracy 0.782, Precision 0.625, Recall 0.225, F1 0.331

The Capital Gain Rule performed best for the chosen business objective because it achieved the highest Precision (0.625) and Accuracy (0.782), making it the strongest baseline.

Error Analysis

False negatives were generally more educated (11.49 years vs. 9.87), worked more hours per week (45.42 vs. 40.61), and were slightly older (43.58 vs. 42.13) than false positives. This shows that relying only on capital gain misses many high-income individuals who have no reported capital gains. The next step is to include more features such as education, hours worked, occupation, and marital status to improve Precision while increasing Recall. Hopefully that will make things more understandable in the future